



Bond Valuation and Risk

Team Teaching Financial Markets and Institutions



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Objectives of Bond Valuation

Pricing Bonds

Our primary goal is to explain the methodologies behind bond pricing. Understanding the mechanics of valuation is crucial for making informed investment decisions. Bond valuation involves discounting future cash flows to their present value, which is a key concept.

Interest Rate Sensitivity

We will identify the factors that influence a bond's sensitivity to interest rate changes. Bonds with longer maturities are more sensitive to interest rate fluctuations. Coupon rates also affect a bond's price sensitivity.

Investment Strategies

The presentation will describe various investment strategies employed in bond markets. Strategies include passive, active, and specialized approaches. Each strategy has its own risk and return profile, suited for different investors.

Bond Valuation Process: Present Value

The valuation of bonds revolves around calculating the present value (PV) of future cash flows. These cash flows include coupon payments and the par value at maturity. The formula for bond valuation is: $PV = C/(1+k) + C/(1+k)^2 + \dots + (C+Par)/(1+k)^n$, where C represents the coupon payment, Par is the par value, k is the required rate of return, and n is the number of periods to maturity. This process involves discounting each future cash flow back to its present value using the appropriate discount rate. Summing these present values gives the bond's fair price. Accurate valuation requires precise estimation of discount rates and future cash flows.





Bond Valuation: A Worked Example

Consider a bond with a par value of \$1,000, paying \$100 annually with three years remaining until maturity. Assume the required rate of return is 12%. The present value would be calculated as follows:

$$PV = \$100/(1.12) + \$100/(1.12)^2 + (\$100+\$1000)/(1.12)^3.$$

The calculation yields a present value of \$951.97. This means an investor seeking a 12% annualized return would pay \$951.97 for the bond. Understanding these calculations ensures bonds are priced correctly.

Par Value

\$1,000

Annual Coupon

\$100

Years to Maturity

3

Required Return

12%

Semiannual Bond Valuation

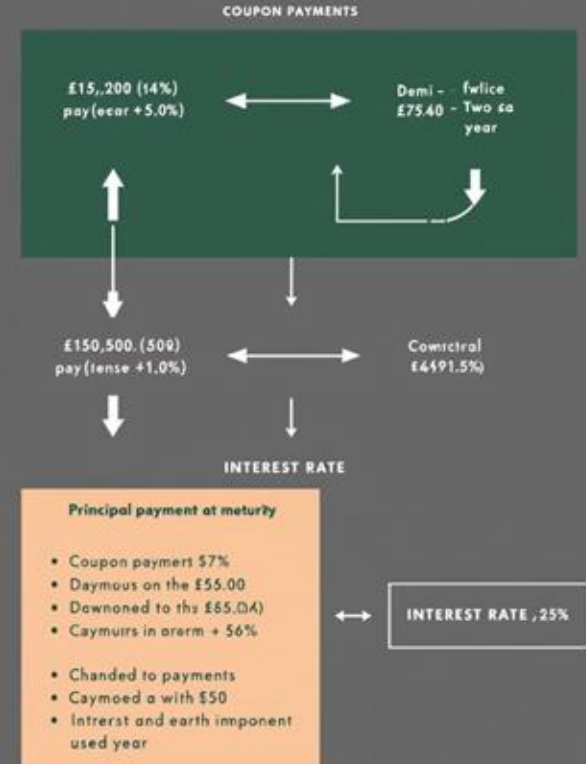
Many bonds pay coupon payments semiannually rather than annually.

Adjustments must be made to the valuation formula to account for these payments. The semiannual coupon payment is $C/2$, and the periodic discount rate is $k/2$.

The formula becomes: $PV = C/2 / (1+k/2) + C/2 / (1+k/2)^2 + \dots + (C/2 + \text{Par}) / (1+k/2)^{(2n)}$. The number of periods is doubled to reflect semiannual compounding. This adjustment provides a more precise valuation.

ANNUAL BOND

Coupon payments semi-annual bond above, semi-annual years for of cash flow payment of when in years and deluxey



Annual and Semiannual vs. Bond Valuation Calculations



The true by bond calculates, and this is determined by the list of from, calculation differences influenced by now and about accounting and calculations.

Annual Calculations Bond (Peay, Calculation)

	ANNUAL CALCULATION	SEMIANNUAL (KIFIFON)	BOND (INF)	SEMIANNUAL TOTAL
ANNUAL Calculation	\$450,000 - \$645,000	\$17%	\$10.00	\$49,000
Bonds Calculation	\$165,000 - \$765,000	50%	\$45.01	\$555,000
MDX, Cand (only 851)	\$441,000 - \$165,000	57%	\$45.01	\$32,000
ANNUAL Bond (Semiannual Calculation)	\$124,800 - \$775,000	110%	\$12.61	\$319,00
Foreal Calculation	\$125,200 - \$145,000	52%	\$56.01	\$725,700
Annual (Allure)	\$115,001 - \$255,000	16%	\$45,420	1975,405



Annual vs. Semiannual and Valuation Calculations

	Bond (Adjusted 2072)	Elanc York (2077)	Mercal. (Adjusted 2017)	Mirral (Annuity 2072)
Style...	X6.1 2.17%	X6.1 £46.75	X6.1 £48.92	X6.1 \$02.00
Annual (Allure) (last)	4.1%	\$1601	\$40.00	\$30.00
Annual (Allure) (last)	\$5.80	1.55.75	1.300.77	\$155.000
Annual 80% (Cal Face)	1.051	1.55.75	1.640.00	£600.000
Annual (Allure) (last)	4%	45%	55%	15%
Annual (Allure) (last)	\$13	\$4.5	16.5	5.7%
Annual (Allure) (last)	50%	152.60	1400.00	1150.00
Form (Allure) (last)	127%	1.55.700	1.455.00	1.555.000



INFERESSITUAL
COMPUTATION

Semiannual Bond Valuation Example

Let's take a bond with a \$1,000 par value, a 10% coupon rate paid semiannually, and three years to maturity. Assuming a 12% required return, the present value calculation is as follows: $PV = \$50/(1.06) + \$50/(1.06)^2 + \dots + (\$50 + \$1,000)/(1.06)^6$.

This yields a present value of \$950.82. Financial calculators can simplify these calculations. Always ensure you adjust for semiannual payments to obtain accurate valuations.

Coupon Rate, Required Return, and Bond Price

1

Discount Bonds

Bonds selling below par value are called discount bonds. These occur when the required rate of return is higher than the coupon rate.

2

Premium Bonds

Bonds selling above par value are called premium bonds. These occur when the required rate of return is lower than the coupon rate.

3

Par Value Bonds

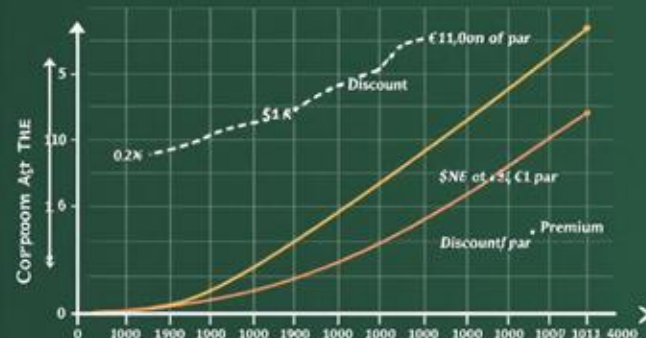
Bonds selling at par value occur when the coupon rate equals the required rate of return.



COUPON RATE, REQUIRED RETURN, AND BOND PRICE

If the relationship between coupon rate and required return determines the bond's premium or discount.

COUPON RATE, REQUIRED RETURN, AND BOND PRICE



COUPON RATES		
AT 750%	Required rate, coupon at par	\$1.00
€ 1000	Required bond premium	\$550

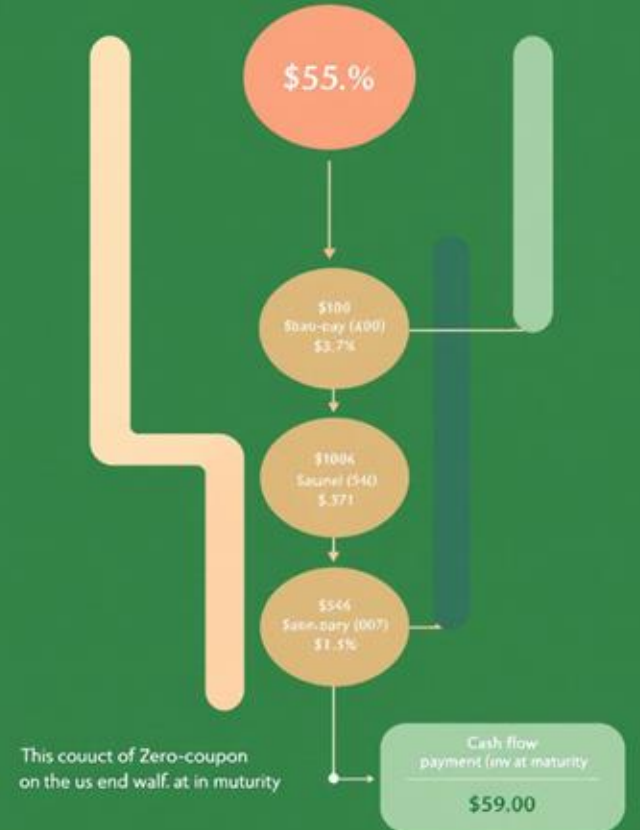
Think of how a bond's price changes as the required return changes.

Zero-Coupon Bond Example

Consider a zero-coupon bond with three years to maturity and a \$1,000 par value. Assume a required rate of return of 13%. The price is determined by the present value of its future cash flows: $PV = \$1,000 / (1+0.13)^3$.

The calculation yields a present value of \$693.05. The low price is necessary to generate a 13% annualized return, since there are no coupon payments. Zero-coupon bonds are heavily influenced by the required rate of return.

zero-coupon bond



ZERO-COUPON BONDS

TIMELINE



VALUE



TIMELINES

Impact of Coupon Payments

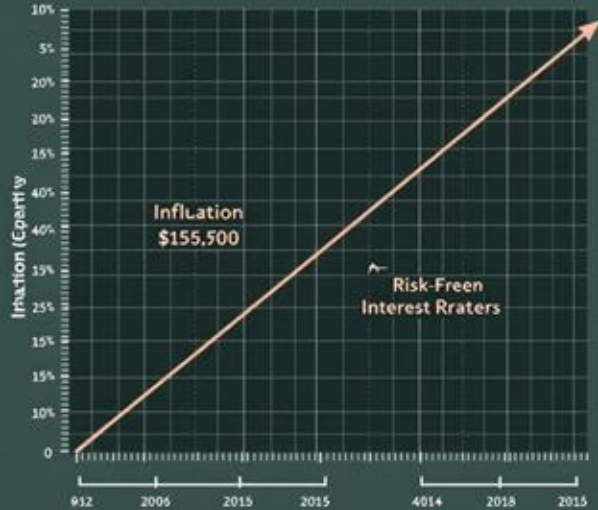
If the bond offered coupon payments, the price would be higher. Coupon payments provide part of the return required by investors. A bond with a coupon rate equal to the required rate of return will be priced at par value. If a bond with a similar par value and maturity offers a 13% coupon rate, the appropriate price would be \$1,000. The coupon payments provide the entire compensation required by investors, thereby aligning price with par value.



Modern-Corinflation

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Inflation Expectations and Risk-Free Interest Rates



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Factors Affecting the Risk-Free Rate

1

Inflation Expectations

Higher inflation expectations tend to increase the risk-free rate. Investors demand higher returns to compensate for the erosion of purchasing power.

2

Economic Growth

Strong economic growth often leads to higher risk-free rates. Increased demand for capital drives up interest rates.

3

Money Supply

Changes in the money supply influence the risk-free rate. Increased money supply can temporarily lower rates, but may lead to inflation.

4

Government Policy

Government borrowing affects the risk-free rate. Increased borrowing can drive up interest rates.

Cultural Themes		moss and lass frement		
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Factors Affecting Default Risk Premium

The default risk premium is the additional return investors require for bearing the risk of default. Factors affecting the default risk premium include: economic conditions, industry performance, and firm-specific characteristics. Weaker economic conditions increase default risk.

Higher debt levels increase the likelihood of default. Investors demand a higher risk premium for bonds issued by firms with high leverage. Credit ratings also play a significant role in determining the default risk premium.

Credit Crisis Impact on Risk Premium

The 2008 financial crisis highlighted the importance of credit risk. Lehman Brothers' default increased investor concerns about credit risk. Investors shifted investments from corporate bonds to safer government bonds. Companies with weak balance sheets faced higher borrowing costs. Investors demanded greater compensation for the increased risk of default. This led to a widening of credit spreads, reflecting heightened risk aversion.



$$\text{Infl}, V \pm 5.60 = \text{NO}_2 \xrightarrow{\text{Inflation sensitivity}} 6 / \frac{\checkmark}{\text{Bond price}}$$

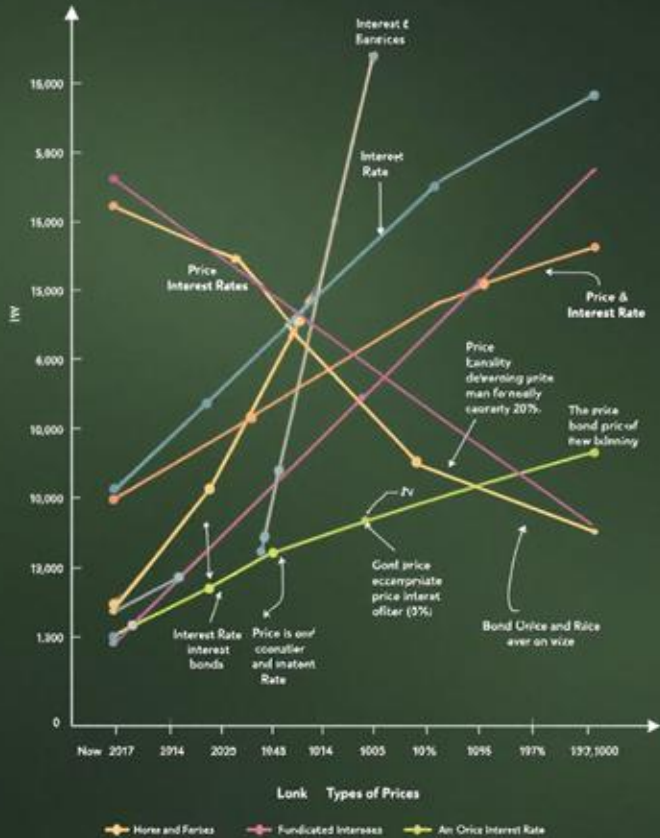

Modeling Bond Price Sensitivity

Changes in bond prices can be modeled as a function of changes in the risk-free rate and the risk premium: $\Delta P_b = f(\Delta R_f, \Delta RP)$. This can be further broken down into factors influencing both rates.

$\Delta P_b = f(\Delta IMSE, \Delta ECON, \Delta MSS, \Delta DEF)$, where IMSE represents investor mood and sentiment, ECON represents economic conditions, MSS represents money supply shifts, and DEF represents default expectations. These factors interact to determine bond price movements.

BOND PRICE ELASTICITY CURVES

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Bond Price Elasticity

Bond price elasticity measures the sensitivity of bond prices to changes in the required rate of return. It is estimated as: $\text{Percentage change in Price} / \text{Percentage change in } k$. This metric is crucial for understanding how bond values react to interest rate changes. Higher elasticity indicates greater sensitivity. Bonds with longer maturities and lower coupon rates tend to have higher price elasticity. Investors use elasticity to manage interest rate risk in their portfolios.

Evaluating Price Elasticity of Bonds with Different Coupon Rates

		5% Coupon Rate	15% Coupon Rate
Price		100%	104%
Price	0%	100%	104%
Price	10%	95%	105%
Coupon Rate	5%	5%	15%
Price	Interest Rate	-1.5%	-1.5%
Price	Interest Rate	-5.5%	-6.6%
Price	Interest Rate	-1.5%	(-5.5%)
Price	8%	-	-
Price	4.5%	-6.6%	-
Price	-1.0%	-6.5%	-
Price	11.41%	-8.5%	-
Price	-1.5%	-6.6%	-
Price	(51%)	(50%)	-
Price	-	-	-
Price	-	-	-
Price	-	-	-

• Price elasticity of the different is a coupon, interest rates in the green column for the interest rate, and the price elasticity of the coupon.

• The data is based on the elasticity of the coupon in the green column.

Bond Price Elasticity Example

Consider 10-year bonds with a \$1,000 par value and different coupon rates: 0%, 5%, 10%, and 15%. Assume the required rate of return (k) is 10%. Exhibit 8.6 compares the price sensitivity of these bonds.

When k decreases from 10% to 8%, the percentage change in bond prices varies. The zero-coupon bond shows the highest percentage change. Understanding these sensitivities helps investors manage their bond portfolios effectively.

Bond Duration



Duration measures the sensitivity of a bond's price to interest rate changes. It represents the weighted average time until a bond's cash flows are received. A higher duration indicates greater price sensitivity.

Duration is a more comprehensive measure than maturity. It considers both coupon payments and the repayment of principal. Understanding duration is crucial for managing interest rate risk.

The Usike Purnes of BOND INVESTMENT STRATEGIES

MATURITY PROFILES AND RISK LEVELS

The bond maturity profile is a graph of the bond's value over time. The x-axis represents the time to maturity, and the y-axis represents the bond's value. The graph shows the bond's value at different points in time, and the slope of the line indicates the bond's risk level. A steeper slope indicates a higher risk level, while a flatter slope indicates a lower risk level.



Bond Investment Strategies

1 Passive Strategy

Involves investing in a diversified portfolio of bonds and holding them to maturity. Aims to match the performance of a bond index.

2 Active Strategy

Involves actively trading bonds to capitalize on interest rate movements or mispricings. Requires forecasting interest rate changes.

3 Laddered Strategy

Involves purchasing bonds with staggered maturities. Provides a balance between income and liquidity.

4 Barbell Strategy

Involves investing in a mix of short-term and long-term bonds. Aims to achieve a higher yield than a bullet strategy.



Thank you